



USA TRIATHLON EDUCATION

LT1, LT2 and VO2 for Cycling

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Overview

Cycling is the most data-driven of the three triathlon disciplines. Power meters, heart rate, and analysis platforms make it easy to locate LT1, LT2, and VO2 effort on the bike and to track how they change.

In cycling, LT2 is Functional Threshold Power (FTP). LT1 and VO2 effort are expressed as power relative to FTP. This guide covers what each marker means on the bike, how to find it, and how to train around it.

LT1 for cycling

LT1 is where lactate first builds above baseline. It is often called the aerobic threshold or Fat Max. Below it, athletes build aerobic base with minimal fatigue, and the limiter is fueling and muscular endurance rather than cardiovascular capacity. Ironman athletes typically ride near or just above it.

LT1 is hard to find without lactate or gas-exchange testing, so most software estimates it from FTP. A practical approach is to find LT2 first, then calculate LT1 as a percentage of it. A typical age-group value is about 70% of FTP, so a 300 W FTP puts LT1 near 210 W.

Athlete category	Typical profile	LT1 as a percentage of LT2
Age-Group	Newer athletes, about 6 to 10 hours per week	About 70%
Performance	High-performance age groupers	About 75%
Elite	Professional athletes (testing is typically done)	About 80%

Use long Zone 2 rides for base, with 40 to 70 mile rides for long-course athletes. Aerobic decoupling is a handy check. Compare power to heart rate in the first and second half of a steady effort longer than about 20 minutes below threshold. Drift under 5% suggests good endurance for that duration and output. Heat, dehydration, and stress also move heart rate, so read it as one window on durability, not a final measurement.

LT2 for cycling: FTP

LT2 is the tipping point where lactate accumulates faster than the body clears it, typically around 3.8 to 4.1 mmol/L. On the bike it is FTP, the power you can hold for roughly 30 to 70 minutes depending on genetics and training. Two protocols dominate:

	20-minute FTP test	Incremental step test
Protocol	Ride all-out for 20 minutes, then take 95% of average power (make sure athletes have been signed off by their doctor for physical exercise, and only complete maximal tests once the athlete is well conditioned to training)	Start low (for example 80 W) and add 10 W every 2 minutes until the athlete cannot hold 80 rpm or more
Strengths	Simple, minimal instruction, works for remote athletes	Own warm-up built in, less painful, shows the heart rate deflection point, RPE at LT2 about 6.5 to 7
Limits	One data point. Very aerobic athletes may be under-estimated, very anaerobic athletes over-estimated	Needs more coaching expertise to interpret. Deflection is harder to see in short-course athletes
Best for	Beginner, new, or remote athletes	Performance athletes or when you need diagnostic data

Breathing changes lag threshold by 1 to 2 steps because lactate takes 3 to 4 minutes to stabilize, so look for steady huffing and puffing sustained for 4 to 5 minutes. Heart rate deflection, where heart rate stops tracking rising power, marks the onset of anaerobic contribution.

FTP versus critical power. FTP is a practical single-number estimate. Critical power is calculated from several maximal efforts at different durations, and it also gives you W-prime, the fixed amount of work you can do above critical power, measured in kilojoules. Think of W-prime as a battery: riding above critical power drains it, and dropping below recharges it. Critical power often sits a few watts above FTP, but the two are not identical.

VO2 for cycling

- **Where it matters:** racing under an hour, particularly short-course and draft-legal athletes who surge repeatedly.
- **Training it:** high-intensity intervals at roughly 95 to 105% of threshold, plus lactate tolerance sessions that alternate sweet spot and VO2 intensities.
- **Lactate at VO2 effort:** typically 8 to 15 mmol/L or more, depending on anaerobic capacity and the muscle recruitment of the sport.
- **Its ceiling:** after the first two to three years of intense training, gains come more from threshold and economy than from further VO2 increases.

FTP alone does not tell the whole story. An athlete can get meaningfully faster without moving FTP for six months by improving fatigue resistance. Look beyond FTP to Functional Reserve Capacity (FRC), fatigue rates, and power after burning a set number of kilojoules.

Long course versus short course on the bike

	Long course (non-draft)	Short course (draft legal)
Demand	Sustained power at or near FTP with minimal fluctuation	Repeated surges above threshold and quick recovery
Key qualities	Fatigue resistance, a high sustainable percentage of FTP	Functional reserve capacity, anaerobic development
Workouts	Threshold-and-hold efforts of 16 minutes or more, hill repeats, Zone 2 base	Threshold intervals with 15-second sprints, lactate tolerance work, spin-ups and pickups
Testing	Sustained efforts or threshold holds	Several durations: 5 seconds, 1, 5, and 20 minutes

A 280 W athlete who can sustain 95% of FTP can outride a 300 W athlete who can sustain only 85%. For long course, build the highest sustainable percentage of FTP rather than chasing absolute FTP. Retest every 3 to 6 months, and in the first week after an adaptation week.